A Single Primitive for Mathematics and Physics: What Distinction Forces, What It Posits, and Where the Boundary Lies

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Abstract

This paper is the synthesis of a program that takes a single primitive, the act of distinction, and asks how far it determines the structures of mathematics and physics. We give the honest map. From one act of distinction, with no axiom of choice, no principle of excluded middle, and no axiom of infinity, the natural numbers, the integers, and a field of ratios are constructed, each with its arithmetic laws proved and the strength of each proof recorded. The canonical reciprocal cost that the physical program rests on is *not* determined by that field of ratios alone: the space of admissible costs is classified and the obstruction is exhibited exactly, as a free choice of orientation on each prime. The continuous completion of the ratio field, the one structure on which the cost becomes forced, is itself *not* a consequence of distinction; it is a separate, named commitment, and the gap is a gap in cardinality, the countable reach of distinction against the uncountable continuum. On that completion, and only there, four algebraic laws force the cost into a single one-parameter family that is exactly the gauge orbit of the canonical cost, and one calibration datum selects the canonical representative uniquely within the orbit. The value of that datum is, at present, a posit rather than a consequence. The resulting picture is stratified, not seamless: mathematics is what distinction permits, physics is what the completion of distinction costs, and the two meet at one named boundary, the passage from the countable ratio field to its uncountable completion. We state the claim at exactly this strength, mark each link as forced, posited, or open, and name the single research result, deriving the calibration unit from the cost of one distinction act, that would convert the cost half of the program from a gauge classification into an unconditional forcing.

Contents

1 Introduction: one primitive, an honest map

A foundational program that proposes to derive both mathematics and physics from a single primitive owes two things to its reader. It owes a precise statement of the primitive, and it owes an honest accounting of how far the primitive actually reaches: what it forces, what it merely permits, what it must posit, and where it runs out. The temptation, in a program of this ambition, is to present an unbroken chain from the primitive to the constants of physics and to call the chain a derivation. We resist that presentation here, because it is not true, and because the truth is both more defensible and more interesting.

The primitive is the act of distinction: the drawing of a difference, the marking of one thing as not another. We write it δ. The claim of the program is not that δ forces everything downstream in one seamless sweep. The claim, stated at the strength we can defend, is a stratification in three layers with one boundary between them.

> *From a single act of distinction, with no axiom of choice, no excluded middle, and no axiom of infinity, the natural numbers, the integers, and a rational field are constructed, each with its arithmetic laws proved and its proof strength audited. The canonical reciprocal cost J is not determined by this rational carrier alone; the space of admissible costs is classified, and the obstruction is exhibited exactly. On the continuous completion, a named commitment that is itself not forced by distinction, shown by a cardinality argument, the four algebraic laws force exactly the one-parameter gauge orbit of J, and a single calibration datum selects J uniquely within it. Mathematics is what distinction permits; physics is what the completion of distinction costs; the two meet at one named boundary.*

Everything in this paper is in service of that paragraph, and nothing in this paper exceeds it. The three technical results it summarizes are developed in full in three companion papers: the construction of the number tower from distinction [1], the proof that distinction does not force the continuum [2], and the classification of the cost as a forced form with a gauge unit [3]. This synthesis states the terminal claim, assembles the three results into a single map, makes the strength of each link explicit, treats the question of inevitability honestly and conditionally, and names the one open frontier whose resolution would strengthen the whole.

1.1 Why stratified, not seamless

The honest map has a shape, and the shape matters. The first layer, arithmetic, is forced: distinction alone, with the weakest possible logical commitments, produces the whole number tower. The second layer, the continuum, is *not* forced: distinction reaches only countably far, and the continuous line that physics requires is a strictly larger object that must be adopted as a separate commitment. The third layer, cost, is forced *only on the completion* and *only up to a gauge*: on the countable ratio field the cost is wide open, on the completion the algebraic laws cut it to a single orbit, and one unit of scale selects the canonical member.

The boundary between layer one and layer three is therefore not a technicality. It is the exact place where the program crosses from mathematics into physics, and it coincides with the place where distinction stops forcing and starts positing. That the two coincide is the central structural fact of the program, and we will show why: the cost is underdetermined on the ratio field for the same reason the ratio field is not complete, namely that a gapped domain insulates local constraints from global ones. Remove the gaps, complete the field, and the same act that fixes the domain also fixes the cost form. The price of removing the gaps is a posit. The honest program names that price.

1.2 What this paper does not claim

Three claims are deliberately absent, and a reader familiar with the program's earlier and more exuberant presentations should note their absence as deliberate.

First, we do not claim that distinction forces the canonical cost J. On the ratio field it does not, and this is a theorem about the ratio field, not a gap waiting to be filled. We claim that distinction forces the cost *form*, the gauge orbit, and that one datum fixes the gauge.

Second, we do not claim that distinction forces the continuum. It does not, and this too is a theorem. The completion is a posit, and we say so in the abstract, in the introduction, and again at the boundary where it is invoked.

Third, we do not claim that the calibration unit, the one datum that selects J within its orbit, is itself derived from distinction. At present it is a posit, a choice of scale. Whether it can be derived is the open frontier of Section 9; until it is, claiming it would be claiming more than is proved.

With these absences marked, the positive content is large, and it is exact. We turn to it.

2 The primitive

2.1 Distinction, and the honest question of how many primitives

The act of distinction is the marking of a difference. To distinguish is to hold two things and judge them not the same. From this single act the program builds everything that follows, and the first duty of an honest account is to say exactly what the act contains.

It contains two moves that we must not blur. There is the act of marking, the production of a new token, and there is the judgment that compares two tokens and returns same or different. The program's strongest slogan would be that these are one primitive, that the comparison is internal to the act. We do not assert that here, because the construction as it stands uses both: an operation that produces a successor token, and a decidable judgment of sameness on tokens. Whether the judgment reduces to the act, so that distinction is genuinely one primitive, or whether recognition is irreducibly the pair, mark and compare, is an open question that we state plainly in Section 9. The honest position is that recognition is at most two primitives and possibly one, and that nothing downstream depends on resolving this, because both the marking and the comparison are available from the start.

What matters for the construction is that the judgment is *decidable*: for any two tokens, same or different is determined, with no appeal to the law of excluded middle as a logical axiom, because the determination is computed, not asserted. This is the first of the strength commitments we track.

2.2 Strength tags

The single discipline that holds the whole program honest is that every result carries a tag recording the strength of commitment its proof requires. We use three tags, in increasing order of commitment.

- **Distinction-only.** The result follows from the act of distinction and finite repetition alone: marking, comparing, and iterating a finite number of times. No completed infinite totality, no choice principle, no nonconstructive case split. This is the weakest and most valuable tag.
- **Trace-closure.** The result requires a completed family of traces, an infinite totality treated as a finished object, such as the set of all finite orbits or a limit of a sequence of them. This is a genuine strengthening over distinction-only, and the completion of the ratio field will carry this tag.
- **Classical extension.** The result requires a classical principle, excluded middle on an undecidable predicate, or a choice principle, used essentially rather than as mere verifier convenience.

The tags are not decoration. They are the content of the honesty claim. When we say arithmetic is forced and the continuum is not, what we mean precisely is that the number tower carries the distinction-only tag throughout, and the completion carries the trace-closure tag and cannot be brought down to distinction-only. The boundary between mathematics and physics in this program is, exactly, the boundary between the distinction-only tag and the trace-closure tag.

3 What distinction permits: the number tower

The first layer of the map is the arithmetic that distinction forces. We summarize the construction and its results; the full development with proofs is the subject of the companion arithmetic paper [1].

3.1 Natural numbers

Iterating the act of distinction produces a sequence of tokens, each marked off from its predecessor. The orbit of a token under repeated marking is a distinction-native natural number; we write the type $\mathbb{N}_\delta$. A successor operation and a zero are immediate, the decidable sameness judgment gives equality, and induction is available because every token is reached from zero by finitely many marks. On $\mathbb{N}_\delta$ one defines addition and multiplication by the usual primitive recursions, and the semiring laws, commutativity, associativity, distributivity, the identities, follow by induction. The order is decidable and agrees with the arithmetic.

Claim 3.1 (Arithmetic of $\mathbb{N}_\delta$, distinction-only)**.** The distinction-native naturals $\mathbb{N}_\delta$ form a commutative semiring with decidable equality and a decidable linear order, and the structure is isomorphic to the standard naturals. Every law is proved under the distinction-only tag: no completed infinity, no choice, no nonconstructive case split.

The content of Claim 3.1 is not that $\mathbb{N}_\delta$ is a novel object; it is isomorphic to the naturals, as it must be. The content is the *tag*. The standard set-theoretic construction of the naturals invokes an axiom of infinity to collect them into a completed set. The distinction-native construction does not: each natural is a finite object, the type is open-ended rather than a completed totality, and no step appeals to a totality of all naturals. That is a strictly weaker commitment, and it is the first thing the program can say that ordinary foundations cannot.

3.2 Integers and ratios

Signed orbits, pairs of natural orbits read as a difference modulo the usual cancellation, give the integers $\mathbb{Z}_\delta$, with the full commutative ring structure and a decidable order extending that of $\mathbb{N}_\delta$, all under the distinction-only tag. Ratio orbits, pairs of a signed orbit and a nonzero natural orbit read as a quotient modulo cross-multiplication, give a field $\mathbb{Q}_\delta$ of ratios, with the field operations, the decidable order, the absolute value, and the reciprocal, again distinction-only. The field $\mathbb{Q}_\delta$ is the distinction-native rational field, and it is isomorphic to the rationals.

Claim 3.2 (The rational field, distinction-only)**.** The distinction-native ratios $\mathbb{Q}_\delta$ form an ordered field with decidable equality and order, isomorphic to the rational field, and every law is proved under the distinction-only tag.

On this field the permission structure of arithmetic, divisibility, primality, the Euclidean algorithm, greatest common divisors, is available and decidable. We call this the *permission* reading of distinction: it tells us what comparisons of orbits are allowed, which orbit divides which, which orbit is prime. Permission is qualitative. It is the mathematics that distinction permits, and it is forced, all of it, under the weakest tag.

Boundary. The number tower stops at $\mathbb{Q}_\delta$. The next object the physical program needs, the golden ratio, is a root of $x^2 = x + 1$, hence involves $\sqrt{5}$, which is not in $\mathbb{Q}_\delta$. So even the first downstream physical quantity already lies outside the distinction-native field. This is the first signal that physics does not live on the ratio field, and it points directly at the boundary of Section 4.

4 Where necessity ends: the continuum is not forced

The second layer of the map is a negative, and it is the hinge of the whole program. We summarize it; the full argument is the companion continuum paper [2].

The physical program needs the continuous real line. It needs it because the cost becomes forced only there (Section 5), because the golden ratio and the other downstream constants live there and not on $\mathbb{Q}_\delta$, and because the limiting and second-derivative notions that physics uses have no traction on a gapped field. The question is whether distinction forces this continuum, the way it forces the number tower. It does not, and the reason is a counting argument that no amount of cleverness in the construction can evade.

Claim 4.1 (The continuum is not forced)**.** The objects that distinction generates form a countable collection: the tokens, the orbits, the signed orbits, the ratios, and any family of these produced by finitely-branching generative rules iterated countably, are all countable. The continuous real line is uncountable. Therefore distinction, whose entire output is countable, cannot force the continuum. The gap between them is a gap in cardinality: the countable reach of distinction against the uncountable line. The completion that closes the gap is a separate commitment, carrying the trace-closure tag, and it is provably stronger than anything distinction-only can reach.

The mathematics behind Claim 4.1 is elementary: it is Cantor's theorem that the reals are uncountable, together with the observation that a generative primitive producing finitely much at each of countably many stages reaches only a countable total. The companion paper makes both halves precise and adds a second, independent argument from model theory, that a countable first-order description of distinction has a countable model, so no such description pins the continuum. The contribution is not the diagonal argument. The contribution is the placement of the boundary: showing that this specific generative primitive, the one the program is built on, cannot reach the continuum, and therefore that the continuum is an addition to the program rather than an output of it.

This is where the program crosses from mathematics into physics, and we mark the crossing precisely. Up to and including $\mathbb{Q}_\delta$, everything is distinction-only, and everything is forced. The completion to the real line is trace-closure, and it is posited. Physics begins at the posit.

Boundary (The completion is named, not derived). We adopt the continuous completion of $\mathbb{Q}_\delta$ as an explicit commitment. We do not derive it from distinction, and we do not pretend to. Every result downstream of this point that requires the completion inherits the trace-closure tag, and we carry that tag forward honestly rather than laundering it into a distinction-only claim.

5 What the completion costs: the cost form forced, the unit a gauge

The third layer is the cost, and it is the load-bearing joint of the entire program, because almost everything in the physical theory downstream flows from the precise form of the cost. We summarize the result; the full treatment, negative first, is the companion cost paper [3], which in turn imports the continuous uniqueness theorem from the functional-equation work [4, 5].

The cost is the canonical reciprocal cost

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1,$$

the price of comparing two magnitudes whose ratio is x. The structural requirements a cost is asked to satisfy are reciprocal symmetry, $F(x) = F(x^{-1})$; normalization, $F(1) = 0$; a composition law governing how costs combine; and, when the domain is continuous, continuity and a unit calibration fixing the local curvature at the unit. The question is whether these requirements force J. The answer splits exactly along the boundary of Section 4.

5.1 On the ratio field, the cost is not forced

Claim 5.1 (Discrete underdetermination)**.** On the distinction-native ratio field, the requirements of reciprocal symmetry, normalization, and the composition law do not force J. The admissible costs are exactly the character costs $F_\chi(u) = \frac{1}{2}(\chi(u) + \chi(u)^{-1}) - 1$, where χ ranges over the homomorphisms of the multiplicative group of positive ratios. Because that group is free abelian on the primes, a character is a free, independent choice of value on each prime, and the admissible costs form an infinite product with one factor per prime. The canonical cost J is the single character given by the inclusion; for every prime there is an admissible cost that agrees with J on all other primes and inverts the orientation of that one.

Claim 5.1 is the honest negative, and it is structural rather than anecdotal: the obstruction is not a handful of counterexamples but the entire character group. A uniqueness statement that recovers J on the ratio field must first fix the cost on every prime, which is to assume agreement with J on a generating set; that is a fact about generation, not a derivation of J from distinction. We never present generator-agreement as forcing.

5.2 On the completion, the cost form is forced

Claim 5.2 (Continuous form forcing and the gauge orbit)**.** On the continuous completion, continuity restricts characters to power maps, and the four laws force the cost into the one-parameter family

$$F_\lambda(x) = \tfrac{1}{2}(x^\lambda + x^{-\lambda}) - 1, \qquad \lambda > 0,$$

which is exactly the solution set, not merely a subset of it. This family is the orbit of J under the group of multiplicative automorphisms of the positive reals, and the action is free and transitive, so the family is a torsor: the residual freedom is exactly one positive real, a gauge coordinate. A single comparison value pins the gauge, and the unit calibration, log-curvature one at the unit, selects the canonical member $F_1 = J$.

The two claims together are the precise sense of "the cost form is forced, the unit is a gauge." Distinction and the algebraic laws force the form, the gauge orbit. One datum fixes the gauge. The value of that datum is a multiplicative-automorphism gauge that distinction does not, by itself, fix. The entire arbitrary content of the cost layer is exactly one positive real number, and everything else about the cost is forced.

Boundary (The calibration unit is a posit, for now). The unit calibration is supplied from outside, as a choice of scale. We do not claim it is derived from distinction. Whether it can be is the open frontier of Section 9.

5.3 Sharpening: the cost form needs only order, not the continuum

Claim 5.2 was stated with continuity, the hypothesis under which a multiplicative endomorphism of the positive reals is a power map. Continuity is stronger than the argument requires. A monotone additive function is linear (Darboux), so a *monotone* multiplicative endomorphism is already a power map, and the form forcing of Claim 5.2 goes through with "continuous" replaced by "order preserving."

The consequence reshapes the boundary. Continuity is about limits, and limits need the gaps of $\mathbb{Q}_\delta$ filled, which is the move to the uncountable line. Monotonicity is about order, and order is already present on the real-algebraic closure of $\mathbb{Q}_\delta$, the countable field in which every distinction-posable polynomial comparison resolves. That field contains $\sqrt{2}$ and $\sqrt{5}$, hence φ, and it is countable. On it the monotone cost forms are already exactly the gauge orbit of J. So the cost form is forced on a countable carrier; the cost layer does not require the continuum.

This does not erase the boundary, it relocates it. The number tower is forced. The cost form is forced, now on a countable real-closed field rather than on the uncountable completion. What still sits on the far side of a non-distinction posit is only this: the genuinely transcendental physical constants the downstream theory names, which use π and the exponential and are therefore not algebraic. Even those land in a countable field, obtained by adjoining their specific values to the rationals, and that field is again a proper subset of the continuum. The continuum is a scaffold for *defining* the transcendental functions, not the home of any answer the program returns. The honest residual posit is the existence of those transcendental values, not the uncountable line itself.

6 The meeting point

We can now state the unification at exactly the strength the three layers support, and not a word stronger.

Theorem 6.1 (The stratified unification)**.** *Let δ be the act of distinction. Then:*

(a) (Mathematics is what distinction permits.) *The number tower $\mathbb{N}_\delta \subset \mathbb{Z}_\delta \subset \mathbb{Q}_\delta$ and its permission structure are forced by δ under the distinction-only tag* (Claims 3.1, 3.2)*.*

(b) (The boundary.) *The continuous completion of* $\mathbb{Q}_\delta$ *is not forced by* δ*; it is a trace-closure commitment, separated from the distinction-only layer by a cardinality gap* (Claim 4.1)*.*

(c) (Physics is what the completion costs.) *On the completion, the cost requirements force the cost into the gauge orbit of* J*, with residual freedom exactly one positive real fixed by one calibration datum* (Claims 5.1, 5.2)*.*

Mathematics and physics meet at one named boundary, the passage from the countable ratio field to its uncountable completion, which is exactly the passage from the distinction-only tag to the trace-closure tag.

The boundary on the cost side. Part (c) is stated over the completion to match the historical form of the uniqueness theorem, but Section 5.3 shows the cost form is already forced over the countable real-algebraic closure of $\mathbb{Q}_\delta$, by order alone. The genuine non-distinction commitment in part (b) is therefore needed only for naming the transcendental constants, not for forcing the cost form. The meeting point is sharper than the uncountable completion: it is the adjunction of specific transcendental values to a countable field.

The meeting point is the heart of the program, and its character is worth stating flatly. The cost is underdetermined on $\mathbb{Q}_\delta$ for the same reason $\mathbb{Q}_\delta$ is incomplete: a gapped domain cannot transmit a local constraint, a curvature at the unit, into a global one, a single cost function, because the gaps insulate the constraint. The act of completing the field, removing the gaps, is the same act that lets the four laws bite and force the cost form. So the boundary where the mathematics ends and the physics begins is not two boundaries that happen to align; it is one boundary seen from two sides. From the side of structure it is the incompleteness of the ratio field. From the side of dynamics it is the underdetermination of the cost. Completing the field closes both at once, at the price of one posit.

This is why the program is stratified and not seamless, and why that is a strength. A seamless chain from δ to the constants of physics would have to hide the posit. The stratified account displays it, localizes it to one place, the completion boundary, and one number, the calibration unit, and proves everything on either side. The physics that the program develops downstream, the golden ratio and the discrete time cadence and the spatial dimension and the numerical constants, all live on the completed side of the boundary, and the flagship's task ends at the boundary that makes them possible. We do not rehearse the downstream derivations here; we locate them, correctly, as living beyond the posit, and we note that even the first of them, the golden ratio, already requires the completion because $\sqrt{5}$ is not a ratio.

7 Inevitability, stated conditionally

A natural further question is whether the distinction core is optional: could a foundation be built that does not contain it? The program has a partial answer, and because the answer is partial we state it as a conditional and resist the temptation to inflate it.

Claim 7.1 (Conditional inevitability)**.** Once an admissible formal foundation is parsed into a common formal-system interface, capturing its tokens, its formation rules, and its judgment of syntactic identity, that foundation contains an embedding of the distinction core: it has marks, it iterates them, and it judges them same or different, which is exactly δ and its decidable comparison. In this sense the distinction core is not an optional ingredient of such a foundation; it is presupposed by the act of writing the foundation down.

The honest qualification is essential. Claim 7.1 is proved for foundations *as parsed into the interface*. It is not yet proved that the major historical foundations, set theory, type theory, category theory, and their variants, each parse faithfully into the interface with their full expressive power preserved. Until at least two such foundations are faithfully parsed and their expressivity established, the inevitability claim is conditional on the parsing, and to present it as unconditional would be to claim inevitability for our own interface, which is circular. We therefore carry the inevitability material as a conditional section of the synthesis rather than as a standalone theorem, and we mark the parsing of real foundations as an open task. The defensible statement is the modest one: any foundation, once written in a system of marks with a decidable identity, already contains distinction, because writing it down is distinguishing.

8 The honest accounting

We collect the strength of every link in one place, so that the reader can see at a glance what is forced, what is posited, and what is open. This table is the program's ledger, and the discipline of keeping it is the program's principal asset.

Link	**Strength**	**Status**
Natural numbers and their arithmetic	distinction-only	forced
Integers and their ring structure	distinction-only	forced
Rational field and its order	distinction-only	forced
Permission structure (divisibility, primality, gcd)	distinction-only	forced
Cost on the ratio field	distinction-only	not forced (classified)
Continuous completion of the ratio field	trace-closure	posited (not forced)
Cost form on the completion (gauge orbit of J)	trace-closure	forced
Selection of J within the orbit	trace-closure	forced by one datum
Value of the calibration datum	trace-closure	posited (open: Section 9)
Inevitability of the distinction core	interface-conditional	conditional
Distinction as exactly one primitive	meta-level	open (Section 9)

Read down the status column. The arithmetic layer is forced, entirely, under the weakest tag. The continuum is posited, and provably so, not forced. The cost form is forced, but only on the posited completion, and only up to a gauge. The gauge is fixed by one datum, whose value is itself posited. Inevitability is conditional on a parsing that is not yet done. And the question of whether distinction is one primitive or two is open. Nothing in this ledger is dressed above its station, and the two genuine open entries are named in the next section as research targets, not hidden as solved.

9 The open frontier

Two questions are open, and one of them, if it closes, upgrades the program's central claim. We state both precisely.

9.1 The move that would force the cost outright

The only free input standing between the imported continuous classification and an unconditional forcing of J is the calibration unit: the requirement that the log-curvature of the cost at the unit equals one. On the completion this is adopted, not derived. The target is to derive it from distinction directly.

Claim 9.1 (Open: calibration from one distinction act)**.** Conjecture: the minimal recognition cost of distinguishing one orbit from its successor, the cost of a single act of distinction, carried to the completion, fixes the second-order coefficient of the cost profile at the unit to exactly one. If this is a theorem, the calibration datum is not a free choice but a consequence of the primitive, the continuous classification runs with no free input, and "distinction forces J on the completion" becomes true without circularity. If it is false, the genuine native curvature is some other value, the forced cost is the corresponding member of the gauge orbit rather than J, and one has learned the true cost of distinction, which is itself a result.

We do not claim either outcome. We mark the question as the single highest-leverage open problem of the program, with a falsifier built in: the one-act cost expansion either has leading curvature one or it does not, and both answers are publishable and more fundamental than the present conditional. The program is built to ship its current, honest claim without this result and to absorb it cleanly if it arrives.

9.2 One primitive or two

The second open question is the one raised in Section 2: whether the same-or-different judgment is derivable from the act of marking, in which case distinction is genuinely one primitive, or whether recognition is irreducibly the pair, mark and compare. Resolving it does not change any downstream result, since both moves are available throughout, but it changes what the program may honestly boast. Either the judgment reduces to the act, and the "single primitive" claim is earned, or it does not, and the honest description is that recognition is the pair. Both are more fundamental than asserting unity without proof, and we leave the question open rather than assert the slogan.

10 Conclusion

The program takes one primitive, the act of distinction, and the honest measure of its reach is a stratification with one boundary. Distinction forces the number tower, all of it, under the weakest logical commitments, with no infinity, no choice, and no excluded middle: mathematics is what distinction permits. Distinction does not force the continuous line that physics requires; the line is a strictly larger object, separated from distinction's countable reach by a gap in cardinality, and adopted as a named posit. On that posited completion, and only there, the cost is forced into the gauge orbit of the canonical reciprocal cost, with exactly one real degree of freedom fixed by one calibration datum: physics is what the completion of distinction costs. The two layers meet at one boundary, the completion of the ratio field, which is simultaneously the end of the distinction-only tag and the beginning of the trace-closure tag, the end of mathematics-as-permitted and the beginning of physics-as-costed.

The strength of this account is precisely its refusal to overstate. It localizes the program's entire posited content to one boundary and one number, proves everything on both sides, treats the inevitability of the core as the conditional it currently is, and names the two open questions as research targets rather than solved problems. A grand claim that arrives with its boundary in the abstract and its open frontier named in the text is a different and sturdier object than a seamless chain that hides its joints. The unification is real, and it is stratified. Distinction permits mathematics; the completion of distinction costs physics; and the single open move that would close the last gap, deriving the unit of cost from the cost of one act of distinction, is stated here as the frontier it is.

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